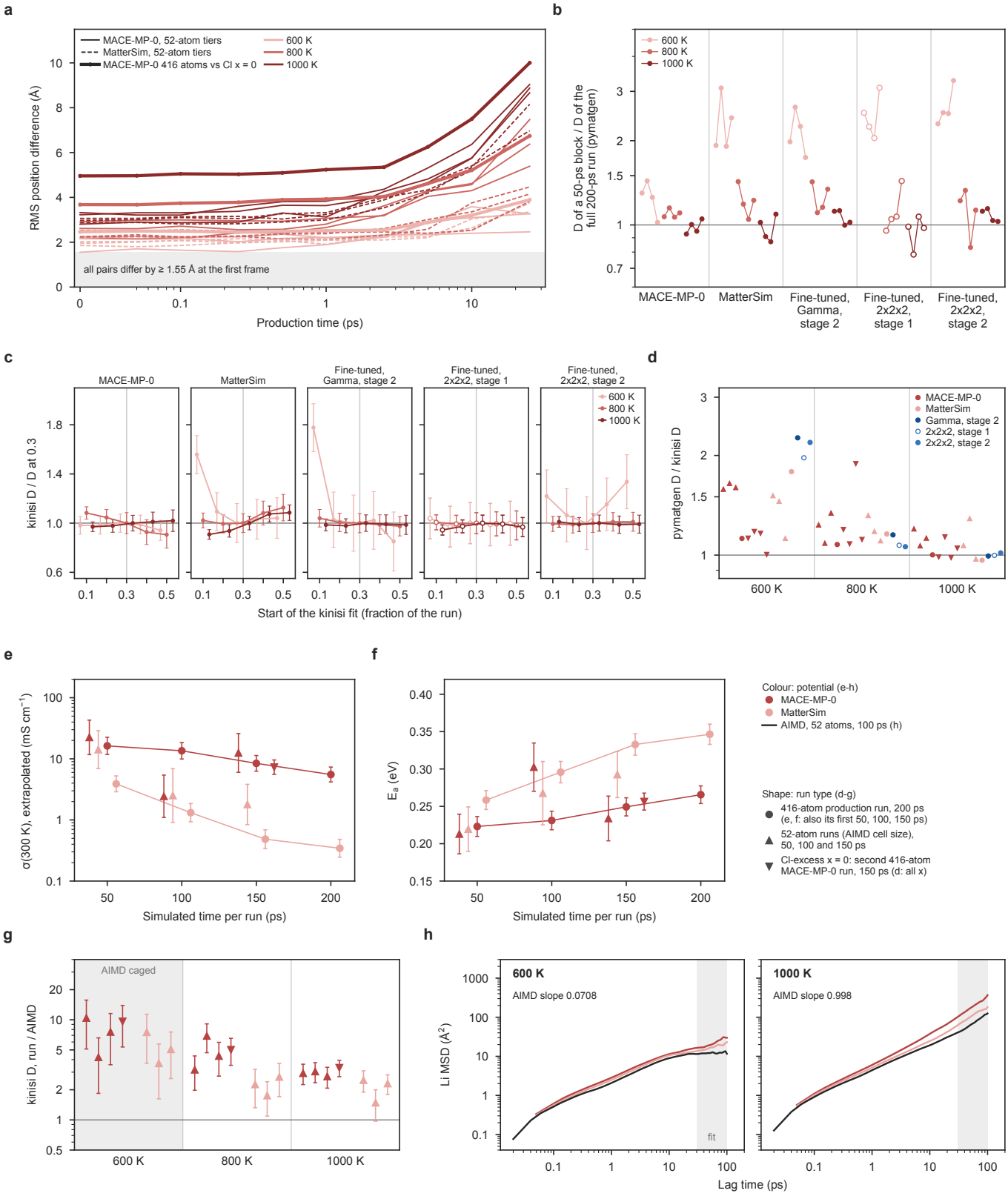

a
MACE-MP-0, 52-atom tiers
MatterSim, 52-atom tiers
MACE-MP-0 416 atoms vs Cl x = 0
600 K
800 K
1000 K
RMS position difference (Å)
all pairs differ by ≥ 1.55 Å at the first frame
Production time (ps)
b
600 K
800 K
1000 K
D of a 50-ps block / D of the full 200-ps run (pymatgen)
MACE-MP-0
MatterSim
Fine-tuned, Gamma, stage 2
Fine-tuned, 2x2x2, stage 1
Fine-tuned, 2x2x2, stage 2
c
MACE-MP-0
MatterSim
Fine-tuned, Gamma, stage 2
Fine-tuned, 2x2x2, stage 1
Fine-tuned, 2x2x2, stage 2
600 K
800 K
1000 K
kinisi D / D at 0.3
Start of the kinisi fit (fraction of the run)
d
MACE-MP-0
MatterSim
Gamma, stage 2
2x2x2, stage 1
2x2x2, stage 2
pymatgen D / kinisi D
600 K
800 K
1000 K
e
σ(300 K), extrapolated (mS cm⁻¹)
Simulated time per run (ps)
f
Eₐ (eV)
Simulated time per run (ps)
Colour: potential (e-h)
MACE-MP-0
MatterSim
AIMD, 52 atoms, 100 ps (h)
Shape: run type (d-g)
416-atom production run, 200 ps (e, f: also its first 50, 100, 150 ps)
52-atom runs (AIMD cell size), 50, 100 and 150 ps
Cl-excess x = 0: second 416-atom MACE-MP-0 run, 150 ps (d: all x)
g
AIMD caged
kinisi D, run / AIMD
600 K
800 K
1000 K
h
600 K
AIMD slope 0.0708
fit
1000 K
AIMD slope 0.998
Li MSD (Å²)
Lag time (ps)